

BIO201 | Anatomy & Physiology I

Course Text

No need to track down a textbook; the readings are built right into this course. We adapted the lessons from the following source:

Betts, J. Gordon, et al. *Anatomy and Physiology*. 2nd ed., Openstax, 2022.

<https://openstax.org/details/books/anatomy-and-physiology-2e>

Course Description

This course explores the foundational aspects of human biology. This course gives students a comprehensive understanding of the structure and function of the human body at both the cellular and organ system levels. By the end of this course, students will be equipped with a solid foundation in anatomy and physiology, paving the way for more advanced studies in the field of life sciences.

Learning Outcomes

After completing this course, students will be able to:

1. Define the basic concepts of anatomy and physiology and list the levels of structural and functional organization of the human body
2. Demonstrate a comprehensive understanding of the building blocks of matter, chemical bonds, chemical reactions, and the role of inorganic and organic compounds in sustaining human life
3. Explain the structure and function of cells and summarize the steps of cell division
4. Describe the characteristics and features of each type of tissue, including their cellular composition, organization, and specialized structures
5. Describe the different structures of the integumentary system and their functions
6. Identify the major bones in the human skeleton, including axial and appendicular skeleton and explain the mechanism of bone growth and development
7. Classify joints and describe their movement and range of motion
8. Summarize the types of muscles of the human body and explain the physiology of muscle contraction
9. Identify the location and state the action of major muscles of the human body
10. Understand the structure and function of neurons and explain the basic principles of neural signaling, including the generation and propagation of electrical impulses
11. Describe the structure and function of the major divisions of the nervous system
12. Explain the structure, function, and sensory perception mechanism of major sensory organs of the human body
13. Identify the major endocrine glands and the hormones they produce and explain the mechanisms of hormone action and the role of the endocrine system in maintaining homeostasis

Course Prerequisites

There are no prerequisites for this course.

Academic Integrity Statement

Academic integrity is the pursuit of scholarly activity in an honest, truthful and responsible manner. Violations of academic integrity include, but are not limited to, plagiarism, cheating, fabrication and academic misconduct. Failure to comply with the Academic Integrity Policy can result in a failure and/or zero on the attempted assignment/examination, a removal from the course, disqualification to enroll in future courses, and/or revocation of an academic transcript.

Course Completion Policy

In order for a course to be considered complete, **all required coursework must be attempted, submitted, and graded.** Required coursework consists of graded assignments. Any Academic Integrity Policy violations may prevent a course from being considered complete.

Assessment Types

StraighterLine courses may include any combination of the assessment types described below. Review the descriptions to learn about each type, then review the Course Evaluation Criteria to understand how your learning will be measured in this course.

Benchmarks

Benchmarks test your mastery of course concepts. You have 3 attempts, and your highest score counts.

Note: Cumulative Benchmarks (final exams) only allow 1 attempt and use a browser-based examination integrity tool.

Capstones

Capstones are project-based assessments that help you apply concepts to real-world scenarios. You have 2 attempts, and your highest score counts.

Checkpoints

Checkpoints are quick knowledge checks on important course concepts. All are open-book, and most have 1-3 attempts.

AI Use-Case Policies

StraighterLine Capstone assessments operate under one of three AI Use-Case Policies. These designations are selected intentionally to support learners in developing digital literacy, ethical reasoning, and authentic communication skills. Each model requires students to engage meaningfully with the course outcomes while adhering to academic standards.

Independent Work Requirement: Capstones with this designation must be completed independently without using AI tools. The goal is for learners to showcase their own understanding and skills without AI assistance. Students are expected to generate and submit original work developed solely through their own reasoning and effort.

AI-Assisted Planning Option: Capstones with this designation may allow AI tools to support brainstorming and assessment planning. If allowed, students will be asked to document any AI assistance by noting how it informed their work. Documentation must be included within the assignment or in a designated reflection field. Examples include describing how an AI tool helped organize an outline, generate ideas, or surface sources for further exploration.

AI-Integration Requirement: Capstones with this designation require AI tools as part of the learning process. Students will be asked to reflect upon their AI interactions and AI contributions to the assessment. Reflections must include which tools were used, how they were used, and what insights students gained from the process. This promotes transparency, ethical use, and metacognitive skill-building.

Course Evaluation Criteria

Your score provides a percentage score and letter grade for each course. A passing percentage is 70% or higher.

There are a total of 1000 points in the course:

Assessment	Points	Learning Outcomes
Benchmark 1: Checkpoints 1-3	125	1, 2, 3, 4
Benchmark 2: Checkpoints 4-6	125	5, 6, 7
Benchmark 3: Checkpoints 1-6	200	1-7
Benchmark 4: Checkpoints 7-9	125	8, 9, 10
Benchmark 5: Checkpoints 10-12	125	11, 12, 13
Benchmark 6: Checkpoints 1-12	300	1-13
Total	1000	

Course Roadmap

This roadmap provides an overview of the checkpoints and lessons covered in this course.

Checkpoint 1: Introduction to Human Anatomy & Physiology

- Basics of Anatomy and Physiology
- Structural Organization of the Human Body
- Anatomical Terms and Body Planes
- Body Cavities and Serous Membranes
- Homeostasis and Its Mechanisms
- Elements and Compounds

Checkpoint 2: Chemistry for Anatomy & Physiology

- Atoms & Subatomic Particles
- Behavior of Electrons
- Types of Chemical Bonds
- Role of Energy in Chemical Reactions
- Characteristics of Chemical Reactions
- Water

- Salt, Acid, and Bases
- The Chemistry of Carbon
- Carbohydrates
- Lipids
- Proteins
- Nucleotides

Checkpoint 3: Cells & Tissues

- Overview of Eukaryotic Cells
- Cell Membrane: Gateway to Cellular Transport and Exchange
- Passive Transport
- Active Transport
- The Cell Cycle
- Control of the Cell Cycle
- Overview of Tissue
- Epithelial Tissue
- Epithelial Tissue: Glandular Epithelium
- Connective Tissue
- Muscle Tissue
- Nervous Tissue

Checkpoint 4: Integumentary System

- Layers of Skin: Epidermis
- Layers of SKin: Dermis and Hypodermis
- Accessory Structures of the Skin
- Functions of the Integumentary System

Checkpoint 5: Bones & The Skeletal System

- Framework of Life: Exploring the Functions of the Skeletal System
- Classification of Bones
- Bone Anatomy
- Bone Cells, Tissue, and Markings
- Histology of Compact & Spongy Bone
- Ossification
- Bone Growth & Remodeling
- Fractures and Bone Repair
- Calcium Homeostasis
- Division of the Skeletal System
- Skull: Braincase Bones Part I
- Skull: Braincase Bones Part II
- SKull: Facial Bones Part I
- Skull: Facial Bones part II
- The Vertebral Column
- Regional Modifications of Vertebrae
- Intervertebral Discs and Ligaments of Vertebral Column
- Thoracic Cage
- The Pectoral Girdle
- Bones of the Upper Limb: Radius, Ulna, and Humerus
- Bones of the Upper Limb: Carpal and Metacarpal
- Pelvic Girdle
- Pelvis

- Bones of the Lower Limb
- Tarsal and Metatarsal Bone

Checkpoint 6: Joints

- Classification of Joints
- Fibrous Joints
- Cartilaginous Joints
- Synovial Joints
- Types of Synovial Joints
- Anatomy of Synovial Joints-I
- Anatomy of Synovial Joints-II
- Types of Body Movements

Checkpoint 7: Muscular System Histology & Physiology

- Overview of Muscular System
- Structure of Skeletal Muscle
- The Neuromuscular Junction and Excitation-Contraction Coupling
- The Sliding Filament Theory
- Energy Sources for Contraction
- Fast and Slow Twitch Muscle Fibers
- Muscle Fatigue, Exercise, and Muscle Performance
- Smooth Muscle
- Cardiac Muscle

Checkpoint 8: Muscular System Gross Anatomy

- Naming the Skeletal Muscles
- Muscles of Facial Expression
- Muscles that Move the Eyes
- Muscles that Move the Lower Jaw and Tongue
- Muscles of the Neck, Head, and Back
- Muscles of the Abdomen and Thorax
- Muscles of the Pelvic Floor and Perineum
- Muscles of Pectoral Girdle
- Muscles that Move the Humerus and Forearm
- Muscles that Move the Wrist, Hand, and Fingers
- Muscles of the Thigh
- Muscles that Move the Feet and Toes

Checkpoint 9: Nervous System I-Basic Structure & Function

- Structural Divisions of the Nervous System
- Functional Divisions of the Nervous System
- Neurons
- Glial Cells
- Functions of Nervous Tissue
- Electrically Active Cell Membranes
- The Membrane Potential
- Graded Potentials
- Synapses
- Neurotransmitter Systems

Checkpoint 10: Nervous System II-Divisions of Nervous System

- The Cerebrum
- The Diencephalon
- The Cerebellum and Spinal Cord
- Blood Circulation and Protective Coverings of the CNS
- The Ventricular System
- Ganglia and Nerves
- Cranial and Spinal Nerves
- Autonomic Nervous System: Sympathetic Division
- Autonomic Nervous System: Parasympathetic Division
- Autonomic Nervous System: Chemical Signaling
- The Structure of Reflexes
- Balance in Competing Autonomic Reflex Arcs
- Central Control
- Sensory Perception
- Sensory Pathways
- Cortical Processing
- Motor Responses II

Checkpoint 11: The Special Senses

- Olfaction
- Gustation
- Visual System
- Hearing and Balance

Checkpoint 12: The Endocrine System

- Overview of Endocrine System
- Hormones
- Hormones of the Thyroid Gland
- Hormones of the Parathyroid Gland
- Hormones of the Adrenal Gland
- Hormones of the Pituitary Gland Part I
- Hormones of the Pituitary Gland Part II
- Pineal Gland
- Organs with a Secondary Endocrine System

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